

Danna Xue

POSTDOCTORAL RESEARCHER

Computer Vision Center (CVC), Barcelona, Spain

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Overview

Currently, I am a postdoctoral researcher in computer vision, especially focusing on image/video editing and low-level vision tasks. I have developed novel approaches to solve different problems ranging from color-related tasks to image segmentation, and have published 11 papers in top-ranked journals and top-ranked conferences. I have research experience in cutting-edge research labs in Spain, China, and Canada, working in high-profile research projects. My research interests mainly lie on: Color (gamut mapping, image/video editing), Low-level vision (image restoration); Image understanding (semantic segmentation).

Work Experience

Universitat Autònoma de Barcelona & Computer Vision Center

Barcelona, Spain

Postdoctoral Researcher in Computer Vision

Jul 2024 - present

- Project 1: High-fidelity color processing in smartphone ISP, e.g. gamut mapping, tone mapping
- Project 2: Content-aware and consistent video editing

Education

Computer Vision Center, Universitat Autònoma de Barcelona | Northwestern Polytechnical University

Barcelona, Spain | Xi'an, China

Ph.D. in Computer Science

Mar 2019 - Jul 2024

- Under joint supervision of **Prof. Yanning Zhang**, **Assoc. Prof. Javier Vazquez-Corral**, and **Dr. Luis Herranz**
- Passed with Cum Laude Mention and supported by the CSC scholarship

Northwestern Polytechnical University

Xi'an, China

M.Sc. in Guidance, Navigation and Control

Sep 2015 - Apr 2018

- Specialised in control theory and astronomical image processing

Northwestern Polytechnical University

Xi'an, China

B.Eng. in Detection, Guidance, and Control Technology

Sep 2011 - Jul 2015

- Specialised in control theory and computer science
- Exchange in Universidad Politécnica de Madrid for 6 months

Publications

CONFERENCE

Bridging the Perception Gap in Image Super-Resolution Evaluation [Project]

Shaolin Su, Josep M. Rocafort, [Danna Xue](#), David Serrano-Lozano, Lei Sun, Javier Vazquez-Corral

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

UDA4Inst: Unsupervised Domain Adaptation for Instance Segmentation [Code]

Yachan Guo, Yi Xiao, [Danna Xue](#), Jose L Gómez, Antonio M López

2025 IEEE Intelligent Vehicles Symposium (IV), 2025

HyperNVD: Accelerating Neural Video Decomposition via Hypernetworks [Project]

Maria Pilligua*, [Danna Xue](#)*, Javier Vazquez-Corral (*Equal Contribution)

International Conference on Computer Vision and Pattern Recognition (CVPR), 2025

PromptNorm: Image Geometry Guides Ambient Light Normalization [Code]

David Serrano-Lozano, Francisco A Molina-Bakhos, [Danna Xue](#), Yixiong Yang, Maria Pilligua, Ramon Baldrich, Maria Vanrell, Javier Vazquez-Corral

International Conference on Computer Vision and Pattern Recognition Workshop (CVPRW), 2025

Burst perception-distortion tradeoff: analysis and evaluation

[Danna Xue](#), Luis Herranz, Javier Vazquez Corral, Yanning Zhang

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023

Slimseg: Slimmable semantic segmentation with boundary supervision

[Danna Xue](#), Fei Yang, Pei Wang, Luis Herranz, Jinqiu Sun, Yu Zhu, Yanning Zhang

ACM International Conference on Multimedia (ACMMM), 2022

JOURNAL

Palette-based Color Harmonization via Color Naming [Code]

Danna Xue, Javier Vazquez-Corral, Luis Herranz, Yanning Zhang, Michael S Brown

IEEE Signal Processing Letters (2024). IEEE, 2024

Take a prior from other tasks for severe blur removal

Pei Wang, Yu Zhu, Danna Xue, Qingsen Yan, Jinqiu Sun, Sung-eui Yoon, Yanning Zhang

Computer Vision and Image Understanding (2024) p. 104027. Elsevier, 2024

Integrating high-level features for consistent palette-based multi-image recoloring [Demo]

Danna Xue, J Vazquez Corral, Luis Herranz, Yanning Zhang, Michael S Brown

Computer Graphics Forum, 2023

Learning depth via leveraging semantics: Self-supervised monocular depth estimation with both implicit and explicit semantic guidance

Rui Li, Danna Xue, Shaolin Su, Xiantuo He, Qing Mao, Yu Zhu, Jinqiu Sun, Yanning Zhang

Pattern Recognition 137 (2023) p. 109297. Elsevier, 2023

Self-supervised monocular depth estimation with frequency-based recurrent refinement

Rui Li, Danna Xue, Yu Zhu, Hao Wu, Jinqiu Sun, Yanning Zhang

IEEE Transactions on Multimedia 25 (2022) pp. 5626–5637. IEEE, 2022

PREPRINT

GLUT: 3D Gaussian Lookup Table for Continuous Color Transformation [Project]

Danna Xue, David Serrano-Lozano, Shaolin Su, Javier Vazquez-Corral

arXiv preprint arXiv:2605.19889 (2026). 2026

Research Experience

Computer Vision Center

Barcelona, Spain

Postdoctoral Researcher

Jul 2024 - present

- **High-fidelity color processing in smartphone ISP**
- Develop an efficient learning-based invertible framework both for gamut reduction and extension.
- Improve the perceptual gamut reduction quality while keeping high fidelity of gamut extension.
- **Consistent video editing with implicit neural representation**
- Accelerate and improve the generalization ability of the layer-based neural video decomposition model with a hypernetwork. (CVPR'25)

York University

Toronto, Canada

Research stay with Prof. Michael S. Brown

Feb 2023 - Apr 2023

- **Palette-based image color manipulation (PG'23, IEEE-SPL'24)**
- Develop an intuitive, user-friendly, and content-aware framework for multiple image color editing.
- Use color-naming for perceptual multi-image color consistency and single image color harmonization.

Computer Vision Center

Barcelona, Spain

Ph.D student with Assoc. Prof. Javier Vazquez-Corral and Dr. Luis Herranz

May 2021 - Jul 2024

- **Perception-distortion tradeoff for multi-image restoration (ICASSP'23)**
- Extend the perception-distortion tradeoff theory by introducing multiple-frame information.
- Analyse the performance of burst image restoration under both aligned bursts and misaligned bursts situations.
- **Slimmable semantic segmentation (ACM MM'22)**
- Propose an effective slimmable semantic segmentation network that can be executed at different capacities during inference.
- Improve the accuracy-efficiency tradeoff by stepwise downward knowledge distillation with the slimmable model.

Northwestern Polytechnical University

Xi'an, China

Ph.D student with Prof. Yanning Zhang

Mar 2019 - May 2021

- **Semantic aware image processing and understanding**
- Propose a self-supervised monocular depth estimation method with semantic guidance. (PR'23)
- Propose a severe blur removal method using semantic priors. (CVIU'24)

Academic Services

Conference Review CVPR 2025 (outstanding reviewer)/26, ECCV 2024/26, WACV 2023, ACM MM 2023/24.

Journal Review Pattern Recognition; IEEE TCSVT; IEEE TGRS

Program Committee Organizer (CVCR&D 2025); Session Chair (CVCR&D 2024); Poster Session Chair (ACMCV 2024).

Skills

Programming Python (PyTorch, Tensorflow, NumPy, Scikit-learn, Pandas, Colour, Matplotlib, etc.), MATLAB.

Miscellaneous Linux, Shell, $\text{\LaTeX}$ (Overleaf/R Markdown), Git.

Language Mandarin (Native), English (Professional, C1), Spanish (Elementary, A1).